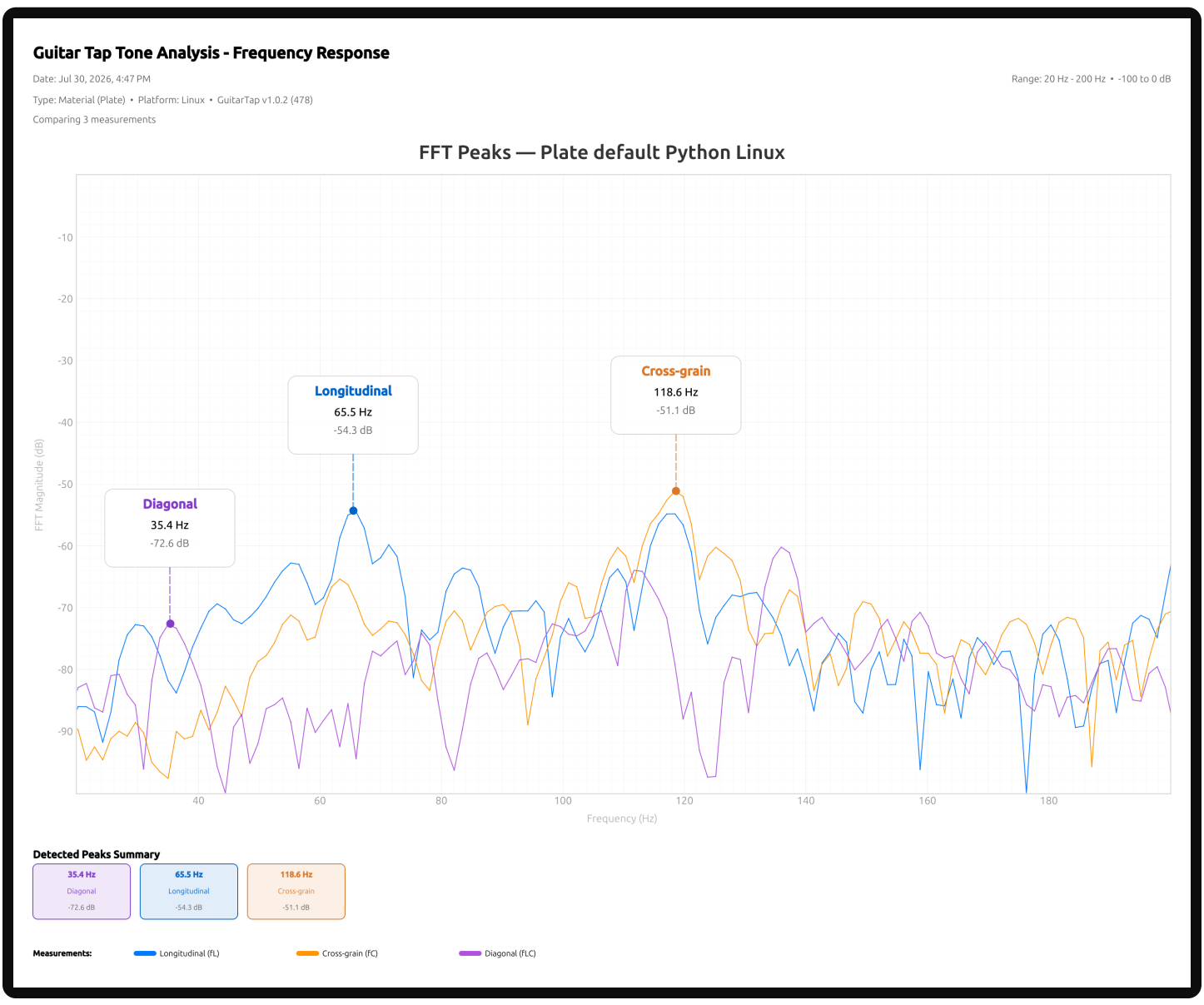
GuitarTap

Tap Tone Analysis Report

Jul 30, 2026, 4:47 PM

Measurement Name: Plate default Python Linux
Type: Material (Plate)
Frequency Range: 20.0 Hz – 200.0 Hz
Microphone: default · uncalibrated

Frequency Spectrum



Detected Peaks

Frequency	Magnitude	Note	Q Factor	Role
35.4 Hz	-72.6 dB	C#1	6.5	Diagonal (fLC)
65.5 Hz	-54.3 dB	C2	12.2	Longitudinal (fL)
118.6 Hz	-51.1 dB	A#2	22.0	Cross-grain (fC)

Sample Dimensions

Length: **557.50 mm** Width: **220.50 mm** Thickness: **4.85 mm**
Mass: **208.0 g** Density: **0.349 g/cm³**

Body Dimensions

Body Length (a): **490 mm** Lower Bout Width (b): **368 mm**
Panel Stiffness: **f_vs = 60 (Classical Top)**

Gore Target Thickness

2.85 mm

Plate Properties

Speed of Sound (L): **4083 m/s**
Speed of Sound (C): **1157 m/s**
Young's Modulus (L): **5.82 GPa**
Young's Modulus (C): **0.47 GPa**
GLC (Shear Modulus): **0.341 GPa**

Specific Modulus (L): **16.7** (Fair)
Specific Modulus (C): **1.3** (Very Good)
Radiation Ratio (L): **11.7**
Radiation Ratio (C): **3.3**

Cross/Long Ratio: **0.080**
typical: 0.04–0.08

Long/Cross Ratio: **12.5**
typical: 12–25

Overall Quality: Good

Three-Tap Measurement Process:

- **1. Longitudinal (fL) Tap**
  Hold plate at 22% from one end along the length, near one long edge (not at the width node). Tap center.
- **2. Cross-grain (fC) Tap**
  Rotate 90°. Hold plate at 22% from one end along the width, near one short edge (not at the length node). Tap center.
- **3. Diagonal (fLC) Tap**
  Hold plate at the midpoint of one long edge. Tap near the opposite corner (~22% from both the end and the side). Measures shear stiffness.

The strongest peak from each tap is auto-selected.